DATE	28-02-2026
TEAM ID	LTVIP2026TMIDS*****
PROJECT NAME	HEMATOVISION: ADVANCED BLOOD CELL CLASSIFICATION USING TRANSFER LEARNING
MAXIMUM MARKS	4 MARKS

7.1 FUNCTIONAL AND PERFORMANCE TESTING :

Functional Testing:

Functional testing is conducted to ensure that the HematoVision system operates according to the defined functional requirements and specifications. The primary goal of functional testing is to verify that each module of the system performs its intended task accurately and reliably.

The testing process begins with validating the user interface components, including image upload buttons, navigation links, and result display sections. It is verified that users can successfully upload microscopic blood cell images in supported formats. The system is tested to ensure that uploaded images are correctly preprocessed, including resizing, normalization, and formatting before being passed to the trained model.

The model loading process is tested to confirm that the saved model file ("blood_cell.h5") is properly accessed and executed without errors. The prediction functionality is verified by comparing predicted outputs with actual class labels from the dataset. Error handling mechanisms are also tested by providing invalid inputs, corrupted files, or unsupported formats to ensure that the system responds with appropriate error messages.

Integration testing is performed to confirm smooth communication between the front-end interface and the back-end server. Overall, functional testing ensures that all features of the HematoVision application work correctly and meet user expectations.

Performance Testing:

Performance testing is a critical phase in the HematoVision project to ensure that the system operates efficiently, reliably, and accurately under different working conditions. Since the application is intended to support medical diagnosis, it must provide fast predictions, stable performance, and high accuracy without system failures.

One of the primary aspects of performance testing is response time analysis. The time taken from image upload to displaying the prediction result is measured carefully. The system must generate predictions within a few seconds to ensure real-time usability in laboratory environments. Slow response time can affect workflow efficiency and user experience.

Another important aspect is model inference time, which refers to the time the trained MobileNetV2 model takes to process an input image and produce a classification result. Optimizing inference time ensures that predictions are delivered quickly without compromising accuracy.

Accuracy performance evaluation is also a key component. The model is tested using unseen test data, and performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix are calculated. These metrics help determine how well the model correctly classifies different types of blood cells and minimizes misclassification.

Load testing is performed to evaluate how the system behaves when multiple images are uploaded sequentially or simultaneously. This ensures that the application can handle increased workload without slowing down or crashing. It verifies system stability during continuous usage.

Scalability testing ensures that the system can manage larger datasets and future expansion. The model's performance is observed when the dataset size increases or when deployed in larger healthcare setups.

Additionally, resource utilization testing monitors CPU, GPU, and memory usage during training and prediction. Efficient resource management ensures that the system runs smoothly even on moderate hardware configurations.

Through comprehensive performance testing, the HematoVision application is validated for speed, stability, scalability, and predictive reliability, making it suitable for real-world healthcare environments where accuracy and efficiency are critical.